THE JUDGE PROTOCOL (TJP)

A Framework for Hardware-Enforced Human Oversight of Artificial Intelligence

AI may analyze, recommend and simulate —
but a human must authorize.

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THE PROBLEM

The Governance Gap Is Global

95%

of simulated nuclear crisis scenarios

AI models recommended escalation. None chose de-escalation. King's College London, 2026.

0

binding cross-border frameworks

The EU AI Act, Bletchley Declaration and Rome Call address particular domains — none mandate runtime human authorization.

2026

Gulf conflict — AI in active operations

AI systems embedded in military targeting without any international governance framework. The gap is live, not theoretical.

In June 2026, a US export directive suspended access to a major model (Fable 5) — restored weeks later, but the episode demonstrated that a single government can unilaterally switch off global AI access. The case for international authorization infrastructure is no longer theoretical.

Pope Leo XIV, Magnifica Humanitas (May 2026): "Not permissible to entrust lethal decisions to AI systems" — the moral call is clear. The enforcement architecture is missing.

AI may analyze, recommend and simulate — but a human must authorize.

WHAT

A universal human authorization layer — every consequential AI action, every domain, every jurisdiction.

HOW

Hardware enforcement below the software stack. No patch, no override — silicon is the guarantee.

WHERE

An open standard. Owned by nobody. Available to every nation, every operator, every chip.

ARCHITECTURE

The Architectural Model: TCP/IP

No packet is delivered without an acknowledgement. No consequential AI action is authorized without a human ACK.

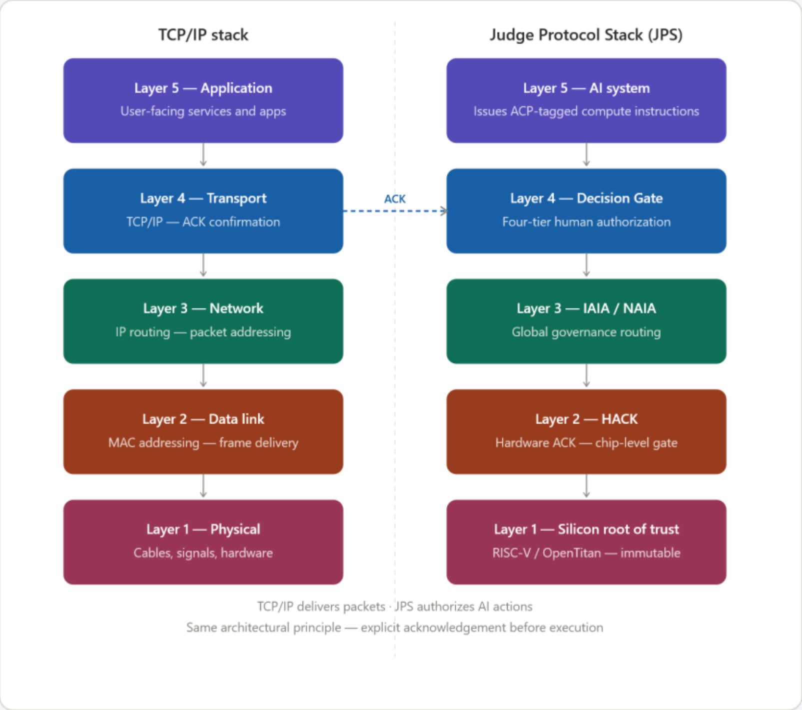


Figure 1 — The Judge Protocol Stack mapped against TCP/IP. Decision Gate at Layer 4; HACK at Layer 2; Silicon root of trust at Layer 1.

DECISION GATE

Four-Tier Authorization

Tier 1

HARD STOP — Existential

Examples: Nuclear · Biological · Chemical · Mass-casualty

Hardware logic. No override. AI cannot recommend, simulate or execute.

HARDWARE HARD STOP

Tier 2

HIGH — Irreversible, Large-Scale

Examples: Military strike authorization · Critical infrastructure shutdown

Multi-party human authorization — two designated authorities required.

MULTI-PARTY ACK

Tier 3

SIGNIFICANT — Material Impact

Examples: Large financial transactions · Clinical treatment decisions · Targeting support

Single designated human authority must authorize.

SINGLE AUTHORITY ACK

Tier 4

ROUTINE — Standard, Reversible

Examples: Automated reporting · Scheduling · Routine analysis

AI executes autonomously — logged and auditable.

AUTONOMOUS + LOG

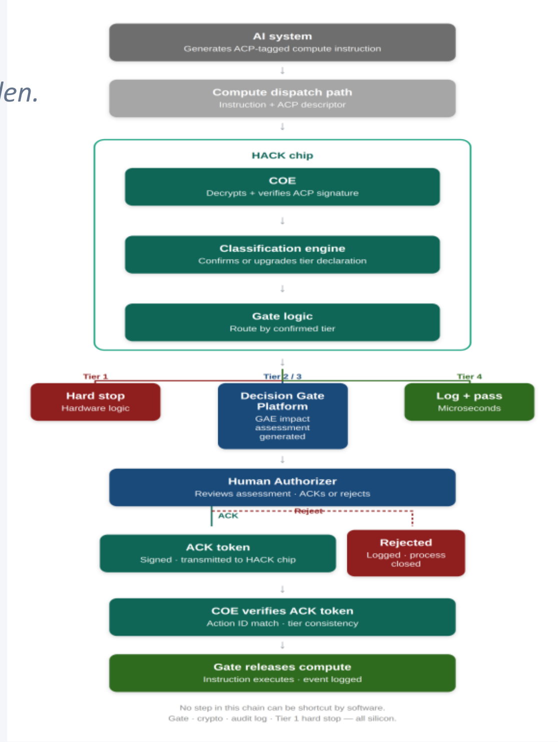
Every Decision Gate event generates an immutable audit log — system ID, action, authorizer, timestamp, rationale, outcome. Retained 10 years minimum.

HARDWARE ENFORCEMENT

The HACK — Hardware ACK

Governance below the software stack — where it cannot be patched, bypassed or overridden.

1. Illustrative TMO chip candidates (e.g. OpenTitan/Ibex) + Chain of Evidence (COE)
2. Tier 1 hard stop in hardware logic — no firmware update overrides it
3. Holds compute capacity gate for Tier 2/3 pending signed human ACK token
4. Tamper-proof on-chip audit log — no software process can alter it
5. Adoption pathway: TPM precedent → ISO/IEC standard → silicon mandate



CERN OHL-S v2 open hardware licence — same institutional role CERN played in the World Wide Web.

HARDWARE ENFORCEMENT

TPM Precedent — HACK Architecture

TPM (today)	HACK equivalent (Judge Protocol)
Application OS, software, services	**AI system** Issues ACP-tagged compute instructions
TPM software stack (TSS) TCG standard API — TPM2-TSS	**HACK TSS equivalent** Open standard API — CERN OHL
LPC / SPI bus interface Physical connection to motherboard	**TMO integration (two paths)** GPU mgmt-processor firmware or motherboard ASIC
TPM chip Crypto engine · key storage · PCR · boot	**HACK module + COE** ACP parser · classification · gate · RISC-V (illustrative)
Endorsement key (EK) Fused at manufacture — never extractable	**Endorsement key (EK)** Fused at manufacture — FIPS 140-3
Silicon root of trust	**RISC-V silicon root of trust**

TPM enforces boot-time security · HACK enforces runtime AI governance · Same silicon trust model, different domain

GOVERNANCE ARCHITECTURE

Three Layers: Silicon → National → Global

Enforceable without being centralized. Universal without requiring any state to surrender sovereignty.

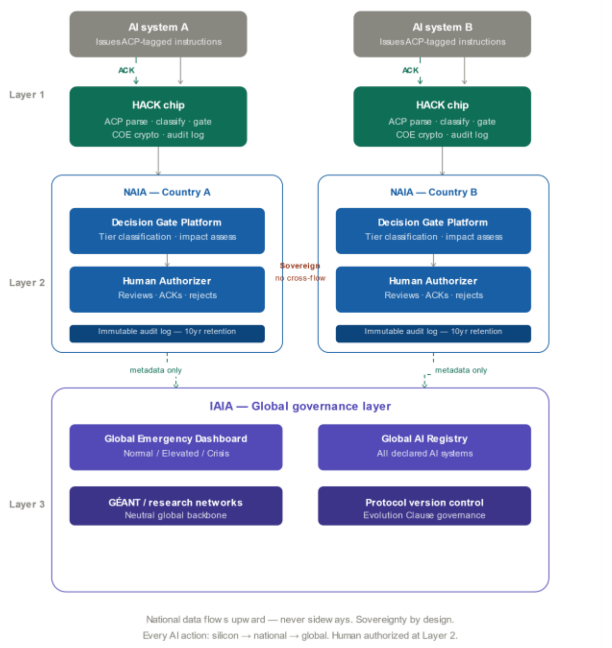


Figure 3 — Every AI action: silicon → national authority → global governance. Human authorized at Layer 2. Data flows upward only — never laterally between nations.

GOVERNANCE ARCHITECTURE

NAIA and IAIA — The Institutional Model

National AI Authority (NAIA)

- Ministerial-level Representative Lead
- Maintains national AI systems registry
- Audits Decision Gate compliance
- Manages AI incidents and enforcement
- Reports compliance metadata to IAIA
- National data remains under national jurisdiction at all times

International AI Agency (IAIA)

- UN specialized agency — modelled on IAEA
- Equal voting rights — no permanent-member veto
- Operates the Global AI Registry
- Global Emergency Dashboard (Normal/Elevated/Crisis)
- Coordinates with IAEA, WHO, BIS/FSB, ITU
- Funded by member states + chip micro-levy at manufacture

No single actor can override the IAIA alert level — it is set algorithmically. Industry contributions carry no voting rights or policy influence.

GOVERNANCE ARCHITECTURE

RG

The Redirect Gate — Refuse and Route

Not a fifth tier — a parallel mechanism. The Decision Gate authorizes an action. The Redirect Gate refuses execution and routes the case to a national authority.

1

Detect

GAE or HACK identifies a Redirect Gate trigger condition, not a standard tier.

2

Refuse Execution

The Decision Gate does not authorize. The action is held in a rejected state.

3

Route

Event and evidence forwarded via the National Response Registry to the national responder.

4

Notify

Human authorizer and NAIA compliance record notified per the use-case SLA.

5

Audit Trail

Event, routing decision, acknowledgement and outcome written to the immutable log.

WALKTHROUGH USE CASE — HUMAN LIFE IMPACT

An AI system detects a threat to human life where self-authorization by the affected individual cannot be relied upon. Execution is refused; the case is routed via the NRR to national health and social services.

Two further use cases follow the same mechanism — Cyber Misuse routes to the national CERT/CSIRT; AI Misuse for Private Gains routes to the relevant sector regulator. Interactive walkthrough: judgeprotocol.org

Sovereignty by Design

National data flows upward to IAIA — never laterally between nations. Each NAIA owns its data.

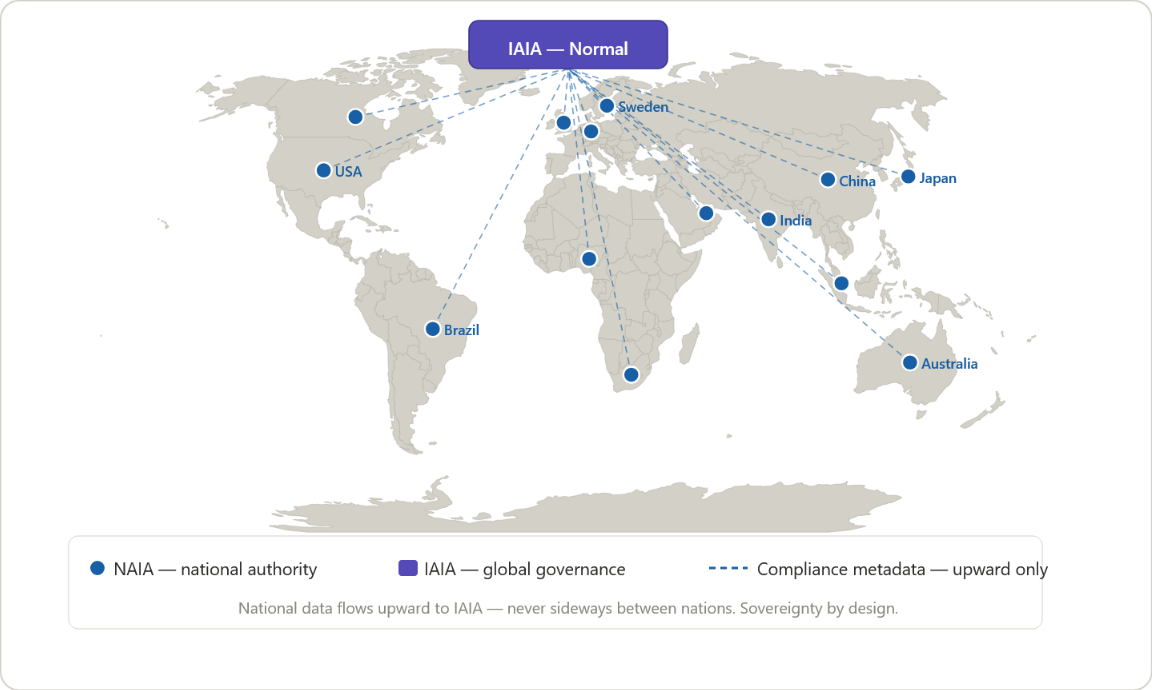


Figure 4 — Global IAIA/NAIA network. Compliance metadata flows upward from national authorities to the IAIA — never laterally between nations.

GOVERNANCE ARCHITECTURE

NAIA Operations Dashboard

● Judge Protocol — NAIA Operations Dashboard ● Normal Live · 14:37 UTC

HACK EVENTS TODAY
48,291
↑ 12% vs yesterday

PENDING AUTHORIZATIONS
7
Tier 2: 2 · Tier 3: 5

HARD STOPS TODAY
0
Last: 14 days ago

LIVE HACK EVENT STREAM

T4	ACK	Case classification — benefit eligibility	SE-TAX-01
T3	ACK	Administrative override — residency status	SE-MIG-04
T2	PEND	Multi-party authorization pending — asset freeze	SE-FIN-09
T3	REJ	Authorizer rejected — deportation order	SE-MIG-07

AUTHORIZATION OUTCOMES

94.1%	4.2%	1.7%
ACK rate	Override	Reject

ACK LATENCY BY TIER

Tier 3 avg	4.2m
Tier 2 avg	2.1h
Tier 4 auto	8µs

Grafana-based · Real-time HACK event stream · Full endorsement verification per tier · 10-year immutable audit retention

IMPLEMENTATION

Four Phases: Prove Before Mandate

1

Phase 1

Sandbox

0–12 months

Reference implementation proven end-to-end across two independent sandbox builds — full Decision Gate and Redirect Gate lifecycle tested. Seeking a founding partner willing to establish and deploy an environment that enables further development toward national/global-level functionality.

2

Phase 2

Reference Implementation

12–24 months

Software stack packaged as open-source Judge Protocol Image, deployable on commodity hardware. IAIA interim secretariat established.

3

Phase 3

Hardened Deployment

24–48 months

TMO integration deployed — GPU firmware or motherboard ASIC path, per manufacturer. Air-gapped appliance under CERN OHL. Full domain rule set coverage across all twelve domains.

4

Phase 4

Silicon Enforcement

48–72 months

HACK core embedded in AI accelerator silicon. Government procurement mandates. Universal silicon-level AI governance enforcement.

The sandbox generates the evidence base that justifies mandatory adoption. Phase 1 asks: one nation, one domain, twelve months.

WHY NOW

The Field Has Arrived at the Gate

Dario Amodei, Anthropic (June 10, 2026)

"Policy on the AI Exponential" — mandatory third-party testing, government power to block unsafe releases, IAEA-style coalition, G7 agenda. Backed by $350M.

Pope Leo XIV (May 2026)

Magnifica Humanitas — "not permissible" to entrust lethal AI decisions. Calls for robust legal frameworks. Rome Call principles now elevated to binding moral obligation.

28 governments (2023)

Bletchley Declaration — multilateral acknowledgement of frontier AI risk. No enforcement mechanism. The gap the Judge Protocol fills.

The Gap

Testing gates the release.

Authorization gates the action.

Every proposed regime lacks a runtime verification layer.

In June 2026, a US export directive suspended access to a major model (Fable 5) — restored weeks later, but the episode demonstrated that a single government can unilaterally switch off global AI access. The case for international authorization infrastructure is no longer theoretical.

The Judge Protocol — published March 2026 — specifies one.

DOI: 10.5281/zenodo.20401666 — timestamped March 2026. Prior art predates the June 10 essay.

EA FRAMEWORK

The Judge Protocol — Enterprise Architecture Framework

Global Layer 3 IAIA	**IAIA** UN specialized agency · IAEA model	**Global Emergency Dashboard** Normal / Elevated / Crisis	**Global AI Registry** All declared AI systems	**GÉANT Backbone** Neutral research network
National Layer 2 NAIA	**NAIA** National authority · every member state	**Decision Gate Platform** Tier classify · impact assess · ACK	**GAE** Governance AI Engine · Claude ref.	**Human Authorizer** Reviews · ACKs · rejects
Silicon Layer 1 HACK	**HACK** Hardware ACK · RISC-V (TBD)	**COE** RISC-V (TBD) · crypto offload	**ACP** Action Classification Protocol	**On-chip Audit Log** Write-once · tamper-proof
Decision Gate Tiers	**Tier 1 — Hard stop** Existential · no override	**Tier 2 — High** Irreversible · multi-party ACK	**Tier 3 — Significant** Material impact · single ACK	**Tier 4 — Routine** Automated · logged

Redirect Gate (escalation to external authorities) and Physical Domain (robots, autonomous systems, weapons hard line) extend this framework — detailed in the full protocol.

AI may analyze | AI may recommend | AI may simulate | ↓ **human must authorize**

Three Open Invitations

1

Founding Sandbox Partner

One national authority. One domain. Twelve months. Deploy the Decision Gate Platform, publish the evidence base, become the founding node of the global network. Sweden is the natural first mover.

2

CERN / Hardware Collaboration

The HACK specification requires a neutral technical authority. CERN played this role for the Web. HLD Sections 4–8 are open for co-development under CERN OHL-S v2 via Openlab.

3

Military Exception Working Group

The most sensitive open question. Must be resolved with ICRC, UN GGE on LAWS, and member state defense establishments before v1.0 ratification. Experts invited.